S^3-Sim: Simulating Humans for Personalized Language Modeling

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Abstract

Developing Large Language Models (LLMs) capable of true personalization remains a significant challenge, primarily due to the scarcity of high-quality, private personalized conversation data. Existing approaches rely on publicly available internet data, which suffers from severe distribution shifts, or synthetic data generated from static personas, which fail to capture the dynamic causal structure of real-world interactions. Consequently, models trained on such data struggle with long-horizon intent understanding and social reasoning. To address this bottleneck, we propose a novel data generation pipeline that synthesizes conversation trajectories by explicitly and structurally maintaining latent user state, encompassing psychological dynamics, beliefs, and evolving social relationships. Unlike naïve scaling strategies, our approach uses these evolving user states as continuous constraints to guide generation, ensuring that the synthesized dialogues reflect realistic causal depth rather than surface-level mimicry. We systematically analyze the impact of different state factors on data quality and demonstrate that our framework enables the controllable scaling of diverse, high-fidelity personalized data. Experiments show that models fine-tuned on our data exhibit superior sample efficiency and significantly improved capabilities in intention inference and theory-of-mind reasoning compared to baselines.

1. Introduction

The development of personalized Large Language Models (LLMs) represents a transition from generic response generation toward agents capable of internalizing individual human contexts (Woźniak et al., 2024). Current research paradigms predominantly focus on system-level orchestration, utilizing external memory wrappers or inference-time context injection to achieve adaptation (Liu et al., 2025; Doddapaneni et al., 2024). However, these modular approaches often fail to address the fundamental challenge of deep personalization: the need for models to internalize the nuanced, state-aware dynamics of human-AI interaction (Qiu et al., 2025a;b). Such internalization is currently bottlenecked by the absence of high-fidelity conversation trajectories that explicitly link linguistic behavior to a user's evolving psychological and social states (Tseriotou et al., 2025; Sun, 2025).

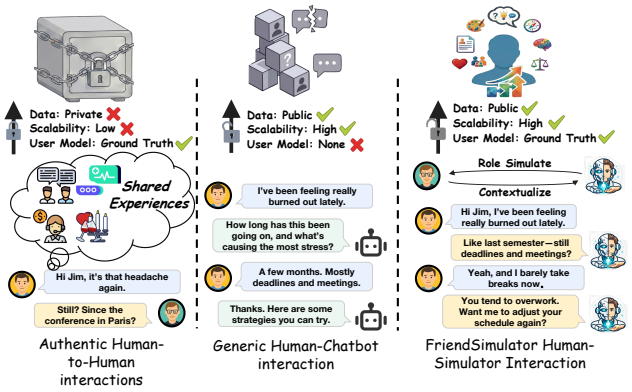


Figure 1. Existing data sources for personalization face a tension between accessibility and depth—real interactions are inaccessible, while prompt-based synthesis scales but remains shallow. Our simulator-based framework (right) bridges this gap by generating scalable, state-aware, deeply personalized trajectories.

This data scarcity arises from a fundamental trade-off between privacy and structural depth (Gumusel et al., 2024). Authentic personalized interactions remain inaccessible due to stringent privacy constraints, while existing synthetic datasets typically rely on stochastic sampling of shallow personas (Jandaghi et al., 2024; Pal et al., 2025) .These methods exhibit a scaling blindness by focusing on surface-level mimicry rather than the causal relationship between a user's internal state and their conversational output (Gui & Toubia, 2023). Because the underlying psychological drivers remain implicit and unmonitored during the generation process, simply increasing data volume fails to improve a model's capacity for complex intention inference or long-term interpersonal reasoning,(Shumailov et al., 2024) leading to low sample efficiency.(Kang et al., 2025; Gan et al., 2026; Dohmatob et al., 2024)

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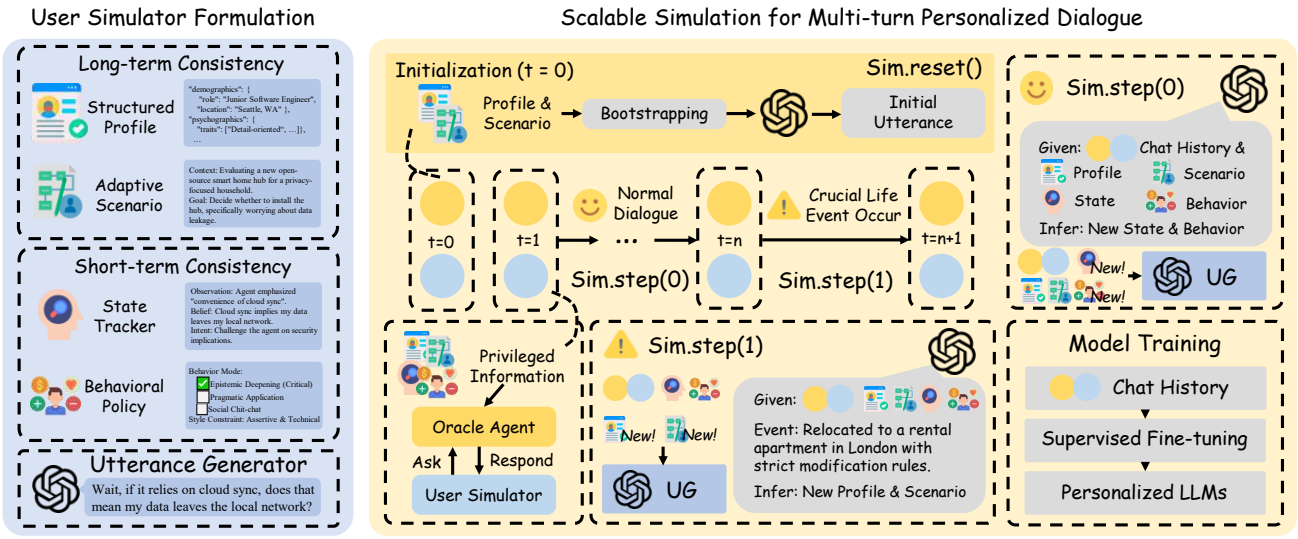


Figure 2. **Overview of S^3-Gen Framework.** The pipeline consists of two primary components: (left) the User Simulator S^3-Sim, which ensures long-term consistency through structured profiles and adaptive scenarios, and short-term consistency via a latent state tracker and behavioral policy; (right) the Scalable Simulation workflow, which includes initial bootstrapping (`Sim.reset()`), incremental turn-based updates (`Sim.step(0)`), and event-driven recursive profile evolution (`Sim.step(1)`). Interaction trajectories are synthesized through an Oracle Agent with privileged access to the latent state, providing high-quality supervision signals for the subsequent Supervised Fine-tuning of personalized LLMs.

To bridge this gap, we introduce S^3 (Structured State Simulation), a framework for State-aware Simulation that explicitly models and maintains a latent, evolvable User State. Our system instantiates S^3-Sim, a simulator that functions as a dynamic state machine with two distinct evolutionary scales: micro-level progression for short-term consistency and macro-level Recursive Hierarchical Update for long-term consistency. By providing ground-truth access to internal variables, including beliefs, emotions, and social relationships (Lin et al., 2023), our platform transforms the simulated user into a fully observable white-box agent as shown in Figure 1. This transparency enables S^3-Gen, a synthesis pipeline that leverages Privileged Distillation to systematically steer conversation patterns and cover data blind spots absent from public corpora. Treating the User State as an explicit constraint allows for the effective scaling of diverse trajectories that authentically reflect the causal complexity of human interaction (Li et al., 2022; Wang et al., 2024b).

Our empirical evaluations demonstrate that data generated through our framework significantly enhances personalized agent performance in SFT settings. The resulting model, S^3-Chat, internalizes the causal complexity of interaction by reverse-engineering the Oracle's decision-making path, achieving superior sample efficiency and theory-of-mind capabilities compared to naïve scaling methods. Our contributions include: (1) S^3-Sim, a scalable simulation environment with explicit state maintenance and recursive hierarchical profile evolution; (2) S^3-Gen, a controllable pipeline for synthesizing state-aware personalized data through Privileged Distillation; and (3) S^3-Chat, a personalized deliverable that demonstrates advanced social intelligence and long-term consistency in language agents.

2. Related Work

Large Language Models Personalization. Personalizing large language models (LLMs) is a central step toward user-adaptive agents.(Tseng et al., 2024; Biancofiore et al., 2025) Most existing approaches rely on external conditioning(Zhang et al., 2024), including prompting and profile augmentation(Wu et al., 2024a; Zhang, 2024; Lyu et al., 2024; Wang et al.), or retrieval-augmented generation(Salemi et al., 2024; Richardson et al., 2023; Shi et al., 2025). These methods typically inject user profiles or interaction histories at inference time(Liu et al., 2025), treating personal context as externally provided and relatively static artifacts.(Li et al., 2024; Magister et al., 2025)

Another line of work pursues parameter-level personalization, including parameter-efficient fine-tuning(Hu et al., 2022), adapters, and user-specific embeddings, which aim to internalize user characteristics into model weights.(Doddapaneni et al., 2024; Chen et al., 2024b) However, it remains limited by data and privacy constraints and still struggles to reliably ground long-horizon user goals.(Salemi & Zamani, 2025; Tran et al., 2025; Kim et al., 2025b)

User Simulation and Modeling. User simulation has long been used to train and evaluate dialogue agents, especially in task-oriented settings. (Sun et al., 2023) Classical simula-

tors often model users as stochastic policies over dialogue acts, driven by predefined goals or agendas.(Schatzmann et al., 2007) This abstraction enables efficient reinforcement learning and controlled evaluation, but it can omit much of the psychological and social realism needed for modern personalized assistants.(Pietquin & Hastie, 2013; Eshky et al., 2012) In many such simulators, the "user" is largely reducible to a goal-completion process, with limited modeling of beliefs, emotions, or interpersonal dynamics (Wang, 2025; Lin et al., 2023).

With the rise of LLMs, a growing body of work uses LLMs as user simulators to generate more naturalistic conversations and to create synthetic training data.(Cho et al., 2024; Sekulić et al., 2024) Compared to agenda-based simulators, LLM-based simulators can produce richer language and broader situational coverage.(Abdullin et al., 2023; Algherairy & Ahmed, 2025) However, many such simulators remain black-box: behavioral drivers are implicit and entangled in the model, making it hard to enforce long-horizon consistency(Davidson et al., 2023; Gui & Toubia, 2023; Mehri et al., 2025), systematically cover rare but critical interaction patterns, or attribute outputs to interpretable factors such as goals or intents.(Guan et al., 2025) Consequently, scaling LLM-simulated interactions may increase surface diversity without guaranteeing deeper structural validity, yielding limited gains for personalization objectives that depend on stable user modeling over time.(Dou et al., 2025)

Latent State Modeling and Intention Inference. Latent state modeling aims to capture unobserved factors that shape user behavior, like beliefs, goals, preferences, and affect.(Purificato et al., 2024; Hasan & Bunescu, 2025) In dialogue research, this is closely tied to dialogue state and belief tracking for maintaining structured estimates of user constraints and intents across turns.(Williams et al., 2016; Mrkšić et al., 2017; Li et al., 2025; Zhang et al., 2025a) Intention inference further targets goal/plan recognition and pragmatic reasoning,(Hoffman et al., 2024; Van-Horenbeke & Peer, 2021; Cao et al., 2025; Shorinwa et al., 2025), and is often evaluated via behavioral benchmarks on social reasoning, theory-of-mind-like understanding, and emotion recognition.(Chen et al., 2024a;c; Sabour et al., 2024) Yet supervision is typically indirect—latent states are rarely observable and are approximated with labels, heuristics, or prompt-specified personas—limiting controllability and causal attribution in long-horizon interactions.(Gui & Toubia, 2023; Subaharan, 2026)

3. Methodology

To bridge the gap between static user profiles and dynamic conversational behaviors, we propose a White-Box User Simulation Framework S^3, as illustrated in Figure 2. Unlike black-box approaches that implicitly model user intent, our framework explicitly maintains a latent User State (S_t) as a causal mediator. This allows us to decompose the personalized data synthesis process into three controllable stages: (1) constructing a diverse scenario space to anchor interactions, (2) generating state-aware user utterances through behavioral decomposition, and (3) synthesizing interaction trajectories via an oracle assistant.

3.1. Problem Formulation

The personalized user simulation is formalized as a Partially Observable Markov Decision Process (POMDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \Omega, \mathcal{R}, \gamma)$. In contrast to standard autoregressive language models that generate tokens based only on surface-level history, this framework posits that observable utterances are realizations of an underlying, evolving latent User State $s_t \in \mathcal{S}$. This state acts as a causal mediator to ensure that linguistic behavior remains consistent with both the psychological attributes of the user and their dynamic intent.

The latent user state s_t at turn t is defined as a structured hierarchical construct:

$$s_t = \langle \mathcal{U}, \mathcal{C}, \mathcal{B}_t \rangle \tag{1}$$

Within this construct, $\mathcal{U}$ denotes the Immutable Profile which includes demographics and core personality traits to establish the static boundary conditions for the persona. The component $\mathcal{C}$ represents the Conversational Scenario, anchoring the interaction in a specific functional context or objective. Finally, $\mathcal{B}_t \in \mathcal{V}_b$ serves as the Dynamic Behavioral Mode, a transient control variable that is determined at the onset of turn t based on the preceding state and history. This component dictates the immediate interaction strategy of the user, such as epistemic deepening, social disclosure, or challenging, which is distinct from the actual linguistic content.

The simulator does not have direct access to the full transition dynamics but perceives the Dialogue History $H_{<t} = \{m_1, a_1, \ldots, m_{t-1}, a_{t-1}\} \in \mathcal{O}$. The simulation process follows a structured causal chain $\mathcal{U} \to \mathcal{B}_t \to m_t$, where the joint generation probability is explicitly factorized into two distinct stages. Initially, in the intent determination stage, the transition function models how the internal intent of the user evolves in response to the history and their profile through a policy defined as:

$$P(\mathcal{B}_t \mid H_{<t}, \mathcal{U}, \mathcal{C}) = \text{Softmax}(f_\phi(H_{<t}, \mathcal{U}, \mathcal{C})) \tag{2}$$

where f_ϕ represents the latent psychological response to the previous action of the assistant. Subsequently, in the linguistic realization stage, once the behavioral mode $\mathcal{B}_t$ is determined, the realization function generates the natural language utterance m_t conditioned on the full latent state

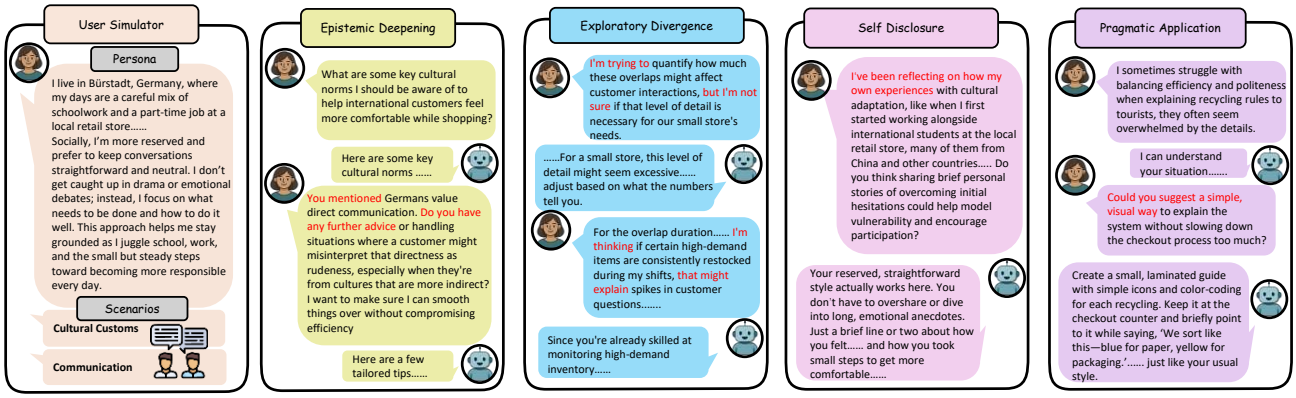


Figure 3. **Selected Examples of Different Behavioral Modes.** Representative interaction strategies including Epistemic Deepening, Exploratory Divergence, Self-Disclosure, and Pragmatic Application are sampled from the behavioral policy $\mathcal{V}_b$.

such that:

$$m_t \sim P_\theta(m \mid \mathcal{B}_t, H_{<t}, \mathcal{U}, \mathcal{C}) \tag{3}$$

By decoupling the strategic intent from the surface-level tokens, the simulation is transformed into a State-Grounded Trajectory Synthesis problem. We instantiate this framework as S^3, a system for State-aware Stateful Simulation, which ensures that synthesized data remains strictly grounded in the latent persona of the user.

3.2. Scenario Space Construction

To prevent the scaling blindness where synthetic data collapses into generic, repetitive tropes, we must ensure the simulator operates over a wide range of realistic scenarios. We construct this scenario taxonomy $\mathcal{C}$ through a dual-strategy expansion pipeline designed to fill the semantic gaps often missing in public datasets while ensuring causal consistency with user attributes.

3.2.1. Dual-Strategy Taxonomy Expansion

We employ two complementary strategies to explore the scenario space, addressing different aspects of coverage and realism. The first strategy, Compositional Self-Expansion, addresses the lack of structural diversity in seed data. By leveraging the internal logic of existing scenarios, we extrapolate to find novel situations that lie "between" or "outside" current clusters. At each iteration, we sample a subset of scenarios as in-context exemplars and prompt the generator to synthesize new categories that combine or extend their elements:

$$c_{\prime 1}, \ldots, c_{\prime n} \sim P_\theta(\cdot \mid \pi_{\text{compose}}, c_{i_1}, \ldots, c_{i_k}) \tag{4}$$

This process forces the model to explore underexplored semantic regions, effectively progressively densifying the coverage of the scenario space.The second strategy, Profile-Grounded Diversification, addresses the issue of causal plausibility. Abstract diversity is insufficient if the scenarios do not align with user capabilities and interests. We explicitly condition scenario generation on the heterogeneity of user profiles. Given a sampled profile $\mathcal{U}$ with attribute set $\mathcal{A}(\mathcal{U})$, we generate scenarios that are naturally relevant to such a user:

$$c_{\prime 1}, \ldots, c_{\prime \ell} \sim P_\theta(\cdot \mid \pi_{\text{profile}}, \mathcal{A}(\mathcal{U})) \tag{5}$$

For instance, rather than a generic "technical issue," a profile for a "privacy-focused security engineer" might trigger a specific scenario like "evaluating local storage vulnerabilities." This ensures that the taxonomy remains grounded in realistic user variance rather than abstract categorical thinking.

3.2.2. Discriminative Filtering

To maintain the quality of the expanded taxonomy, we employ a discriminator to filter candidates based on three criteria: abstraction, distinctness, and validity. **Abstraction Assessment** ensures that scenarios represent generalizable interaction categories (e.g., "career planning") rather than overly specific instances (e.g., "applying to Company X"), allowing for diverse downstream instantiation. **Distinctness Evaluation** prevents semantic redundancy by computing embedding-based similarity scores between candidates and the existing set $\mathcal{C}$, rejecting any candidate c' where $\max_{c \in \mathcal{C}} \text{sim}(c', c) \geq \theta_{\text{sim}}$. Finally, **Structural Validity Checking** acts as a sanity check to remove malformed or nonsensical outputs, ensuring all retained scenarios are semantically coherent and relevant to the simulation goals.

3.3. Stateful User Simulation S^3-Sim

A static user profile is insufficient for evaluating long-term personalization, as real human contexts are non-stationary. To capture the complexity of non-stationary human contexts, we develop S^3-Sim, a simulator that operates as a dynamic state machine with two distinct evolutionary scales: micro-level progression driven by the conversation itself, and macro-level evolution driven by external life events.

3.3.1. Dialogue-Driven Progression

In the absence of external shocks, the user state s_t evolves through the natural accumulation of conversation history and shifting intentions. We term this Dialogue-Driven Progression. While the core profile $\mathcal{U}$ and scenario $\mathcal{C}$ remain constant within a session to anchor the interaction, the latent state s_t is updated at each turn by conditioning on the previous state s_{t-1} and history $H_{<t}$. This step-wise transition allows the model to maintain short-term consistency by tracking evolving beliefs and transient emotional valences. To prevent the collapse into passive compliance often seen in RLHF models (Ouyang et al., 2022), we explicitly model diverse interaction modes $\mathcal{B}$. At each turn, a new behavioral mode $\mathcal{B}_t$ is sampled from a variance-maximizing distribution conditioned on the updated state:

$$\mathcal{B}_t \sim P_{\text{policy}}(\cdot \mid s_t, H_{<t}, \mathcal{U}, \mathcal{C}) \tag{6}$$

For instance, a user might transition from Epistemic Deepening (b_1) to Pragmatic Application (b_4), or suddenly shift to Memory Verification (b_5). This ensures that even within a fixed context, the user exhibits diverse, non-monotonic trajectories.

3.3.2. Event-Driven Context Evolution

To handle concept drift where user preferences mutate, we introduce an event variable E_t for macro-level evolution. Following the structural paradigm of DeepPersona (Wang et al.), we organize the user profile $\mathcal{U}$ as a narrative-complete hierarchical JSON taxonomy with twelve primary attribute domains.

When a significant event E_t occurs, a reasoning module executes a recursive modification protocol, traversing the tree from root to leaves. Each node update is explicitly conditioned on parent attributes to ensure contextual integrity. To prevent personality collapse, we implement a *Backtracking Validation* step grounded in semantic stability. We define the persona drift as the cosine distance between profile centroids:

$$D = 1 - \cos(\phi(\mathcal{U}), \phi(\mathcal{U}')) \tag{7}$$

The recursive transformation is modeled as:

$$\mathcal{U}' = f_r(\mathcal{U}, E_t) \quad \text{s.t.} \quad D < \tau \tag{8}$$

where τ is a stability threshold that ensures core persona anchors remain stable, f_r denotes the recursive modification protocol that traverses the hierarchical profile $\mathcal{U}$ to incorporate event E_t while maintaining parent-child contextual integrity. The function $\phi(\cdot)$ maps the narrative-complete persona into a latent vector space (i.e., all-MiniLM-L6-v2), allowing the drift D to be quantified as the semantic distance between the original and updated profile states. This update ensures that subsequent user behavior aligns organically with the updated reality.

3.3.3. State-Grounded Utterance Realization

Once the state is determined—either through step-wise dialogue updates or deferred recursive event updates—the realization function generates the final utterance. Crucially, we employ Raw Attribute Injection: rather than compressing the profile $\mathcal{U}$ into a summary, we provide the generator with the full, updated attribute set $\mathcal{A}(\mathcal{U}')$ from the JSON tree. The instruction "Given the updated background $\mathcal{U}'$ and current intent $\mathcal{B}_t$, express yourself naturally" ensures that the generated language is strictly causally consistent with both the micro-level intent and the macro-level context constraints established during the recursive update.

Algorithm 1 Dual-Scale Update and Distillation

Require: Profile tree $\mathcal{U}$, Models P_o, P_ϕ
Ensure: Dataset $\mathcal{D}$

while simulating **do**
  $\mathcal{C} \leftarrow$ GenerateScenario($\mathcal{U}$) {Regenerate after deferred macro-update}
  $H \leftarrow \emptyset, s_0 \leftarrow \langle \mathcal{U}, \mathcal{C}, \emptyset \rangle$
  repeat
    $s_t \leftarrow$ UpdateShortTerm(s_{t-1}, H) {Turn-based incremental update}
    $\mathcal{B}_t \sim P_{\text{div}}, m_t \sim P_{\text{gen}}(s_t, \mathcal{B}_t)$
    $a_t \sim P_o(H \cup \{m_t\}, s_t)$ {Oracle accesses privileged state}
    $H \leftarrow H \cup \{m_t, a_t\}, \mathcal{D} \leftarrow \mathcal{D} \cup \{(H, a_t)\}$
  until session ends
  if event E occurs **then**
    RecursiveUpdate($\mathcal{U}.root, E, \emptyset$) {Deferred recursive evolution}
    $D \leftarrow 1 - \cos(\phi(\mathcal{U}_{old}), \phi(\mathcal{U}))$ {Quantify persona drift}
    if $D \geq \tau$ **then**
      $\mathcal{U} \leftarrow \mathcal{U}_{old}$ {Rollback to maintain core identity}
    end if
  end if
end while

function RecursiveUpdate(n, E, ctx)
  $n.attr \leftarrow$ LLM(n, E, ctx) {Conditioned on all parent attributes}
  for each $child$ in $n.children$ **do**
    RecursiveUpdate($child, E, \{ctx, n.attr\}$)
  end for
end function

3.4. Interaction Synthesis via Privileged Distillation

To bridge the gap between surface-level mimicry and deep social reasoning, we implement S^3-Gen, a multi-turn interaction synthesis pipeline grounded in the Learning Using Privileged Information (LUPI) paradigm (Vapnik & Iz-

mailov, 2015). This framework treats interaction synthesis as an asymmetric teacher-student paradigm where an Oracle Assistant, acting as the teacher, possesses full visibility into the latent user state s_t. Unlike standard agents limited to the dialogue history $H_{<t}$, the Oracle leverages a structured state maintained as a dynamic JSON object containing belief sets and emotional valences. This privileged access allows the Oracle to generate responses grounded in the causal drivers of user behavior, effectively mapping high-dimensional latent dynamics into observable linguistic signals.

Algorithm 2 Privileged Distillation Loop (S^3-Gen)

Require: Profile $\mathcal{U}$, Scenario $\mathcal{C}$, Models P_o, P_ϕ
Ensure: Distilled dataset $\mathcal{D}$
1: Initialize state $s_0 = \langle \mathcal{U}, \mathcal{C}, \emptyset \rangle$ as structured JSON
2: **for** $t = 1$ to T **do**
3: Sample mode $\mathcal{B}_t \sim P_{div}(\mathcal{B} \in \mathcal{V}_b)$ to ensure interaction variance
4: User generates $m_t \sim P_{gen}(m \mid s_t, \mathcal{B}_t)$
5: $s_t \leftarrow \text{UpdateShortTerm}(s_{t-1}, m_t)$ {Refine beliefs based on latest utterance}
6: Oracle computes expert response $a_t \sim P_o(a \mid H_{<t}, s_t)$
7: Compute $\mathcal{L}_d$ and update student parameters ϕ
8: Update $H_{t+1} \leftarrow H_t \cup \{m_t, a_t\}$ and evolve state s_{t+1}
9: **end for**

The core of this process involves training a student model P_ϕ to internalize the user modeling capabilities of the Oracle P_o without direct access to s_t during inference. Formally, we optimize the student model to match the Oracle's decision-making logic by minimizing the negative log-likelihood (NLL) over the privileged trajectories. The distillation objective is defined as:

$$\mathcal{L}d(\phi) = -\mathbb{E}(H_{<t}, a_t) \sim \mathcal{D}[\log P_\phi(a_t \mid H_{<t})] \quad (9)$$

By learning to predict the Oracle's privileged actions a_t solely from the observable history $H_{<t}$, the student is compelled to develop internal representations that implicitly reconstruct the underlying latent variables s_t. This transformation provides a high-fidelity supervision signal that guides the student toward sophisticated intention inference and theory-of-mind reasoning, effectively bypassing the need for explicit state tracking at runtime.

To mitigate model collapse where synthesized data becomes repetitive, we explicitly control the distribution of the behavioral mode $\mathcal{B}_t$. We define diverse labels, including epistemic challenge and social conflict, which are sampled from a prior distribution designed to maximize trajectory variance. Furthermore, the evolution of s_t is governed by a conflict detector that triggers an event-driven update whenever a response contradicts structural user beliefs.

The synthesis of interaction trajectories via S^3-Gen facilitates a robust data distillation process. Following the collection of privileged trajectories $\mathcal{D} = \{(H_{<t}, a_t)\}$, we perform Supervised Fine-tuning (SFT) on a pre-trained base model to instantiate S^3-Chat. During this stage, the student model is strictly limited to the observable dialogue history $H_{<t}$ and remains agnostic to the privileged state s_t used by the Oracle. By minimizing the distillation loss $\mathcal{L}_d$, S^3-Chat is compelled to reverse-engineer the Oracle's decision-making path, thereby implicitly reconstructing the latent user dynamics and behavioral patterns from surface-level linguistic cues. This deliberate information asymmetry ensures that the final model develops an advanced capacity for Theory-of-Mind (ToM) and intent recognition, allowing S^3-Chat to maintain high personalization quality during inference without requiring access to unobservable user metadata.

4. Experiments

Our experiments aim to systematically evaluate the S^3 framework's capacity to foster deep personalization and sophisticated social reasoning in language agents. We seek to determine whether internalizing latent user dynamics through state-aware simulation and privileged distillation allows S^3-Chat to surpass the limitations of surface-level mimicry prevalent in current models. The evaluation is designed to investigate several core scientific objectives: first, we assess whether our state-internalization approach enables superior long-term memory and preference adherence compared to models utilizing external memory wrappers or standard supervised fine-tuning; second, we examine the extent to which the asymmetric Oracle-Student paradigm enhances the model's theory-of-mind capabilities, particularly in reasoning about nested mental states and social intentions; finally, we explore the sample efficiency of our causally-grounded synthetic trajectories, testing if high-quality data can enable open-source models to rival significantly larger commercial systems in complex personalized interactions. To this end, we conduct a comprehensive evaluation across seven benchmarks spanning long-term memory, psychological inference, and goal-driven social dynamics.

4.1. Experimental Setup

Benchmarks. We evaluate our approach on seven benchmarks spanning three core capabilities essential for personalized agents:

(1) **Long-term Memory & Personalization.** To assess the ability to track user states over time, we use *PersonaMemV2* (Jiang et al., 2025) across 20–60 sessions with context lengths maximum 128k tokens, and *LongMemEval* (Wu et al., 2024b), a scalable benchmark for long-term interactive memory that provides standardized long-history settings (e.g., $\sim$115k tokens per instance and an extended setting

with hundreds of sessions reaching ∼1.5M tokens). Finally, *PrefEval* (Zhao et al., 2025) specifically targets preference recognition, evaluating whether models can infer and adhere to user choices across diverse topics.

(2) **Theory of Mind.** We employ *ToMi* (Le et al., 2019) to evaluate foundational false belief understanding (first- and second-order). For more complex social inference, we use *BigToM* (Gandhi et al., 2023), which extends evaluation to large-scale scenarios requiring fine-grained belief attribution, goal inference, and action prediction.

(3) **Social Interaction & Emotional Understanding.** We assess goal-driven interaction quality using *Sotopia* (Zhou et al., 2023), measuring dimensions such as goal completion and social appropriateness in complex interpersonal dynamics. Furthermore, *EQBench* (Paech, 2023) is used to evaluate emotional alignment by requiring models to predict emotional intensity in complex social scenarios.

Baselines. We compare against four categories of approaches: **(1) Closed-source Models**, GPT-4o-mini and GPT-5-mini, as representatives of state-of-the-art commercial systems that serve as upper-bound references, representing the capabilities achievable through massive-scale pretraining and RLHF without specialized personalization training; **(2) Open-source Base Models**, Qwen2.5-7B-Instruct (Qwen et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024), as open-source baselines that serve as direct comparisons and as the foundation for our fine-tuning experiments; **(3) Memory-Augmented Methods**, *Mem0* (Chhikara et al., 2025), which implements a hierarchical memory layer that stores and retrieves user information across sessions with multi-level memories (user, session, agent) and semantic retrieval, demonstrating strong performance on long-term memory benchmarks while reducing context length by approximately 90%; and **(4) Specialized Reasoning Methods**, *Sotopia-π* (Wang et al., 2024a), *AutoToM* (Zhang et al., 2025b), and *Thought Tracing* (Kim et al., 2025a), which respectively improve social intelligence through iterative self-play and feedback, implement automated theory-of-mind reasoning through Bayesian inference over mental state models, and introduce hypothesis-driven theory-of-mind reasoning that explicitly traces thought processes to understand mental states.

Implementation Details. We implement the S^3-Gen pipeline using GPT-4o-mini as the backbone for both the Oracle Assistant and the User Simulator S^3-Sim. The privileged interaction dataset $\mathcal{D}$ consists of 1k multi-session dialogues, totaling approximately 20 turns, where each trajectory is annotated with the latent user state s_t during the synthesis phase. For the distillation process, we fine-tune Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct as the student models to instantiate S^3-Chat.The training is conducted via a standard Supervised Fine-tuning (SFT) procedure where the model is optimized to predict the Oracle's responses a_t conditioned solely on the dialogue history $H_{<t}$, thereby enforcing the implicit reconstruction of the hidden state s_t. We employ the AdamW optimizer with a learning rate of 2e-5, a linear warmup over 3 of the total steps, and a cosine learning rate scheduler. The models are trained for 1 epoch with a global batch size of 128 on 8×H100 80GB GPUs. To ensure reproducibility, we use greedy decoding with a temperature of 0 for all benchmark evaluations. For the hierarchical profile tree $\mathcal{U}$ in S^3-Sim, we follow DeepPersona (Wang et al.) and utilize a structured JSON format with a maximum depth of 3 levels to maintain manageable complexity during recursive updates.

4.2. Main Results

Table 1 presents our main evaluation results across all seven benchmarks. We report accuracy for classification tasks (PersonaMemV2, LongMemEval, ToMi, BigToM, PrefEval), EQ score for EQBench, and average goal achievement score for Sotopia.

Table 1 compares *our method*, S^3-Chat (w/ Qwen2.5-7B), with closed-source models, open-source instruction-tuned baselines, and representative prior methods across seven benchmarks that evaluate long-term memory, theory-of-mind (ToM) reasoning, social interaction, emotional intelligence, and preference alignment.

Overall, S^3-Chat achieves the strongest performance among the reported open-source systems across most benchmarks, while remaining competitive with substantially larger closed-source models. On long-term memory benchmarks, S^3-Chat improves over its backbone Qwen2.5-7B on both PERSONAMEMV2 (0.5467 vs. 0.5263) and LONGMEMEVAL (0.1660 vs. 0.1500), and also surpasses the memory-augmented baseline Mem0 on LONGMEMEVAL (0.1660 vs. 0.1620). These results suggest that S^3-Chat better supports persistent persona modeling and long-horizon information retrieval.

The largest gains are observed on ToM reasoning. S^3-Chat attains a near-ceiling score on TOMI (0.9489), outperforming all baselines by a substantial margin, including specialized ToM methods such as AutoToM (0.8123) and Thought Tracing (0.6600). On BIGTOM, S^3-Chat remains competitive (0.7007), outperforming several prior methods (e.g., Sotopia-π: 0.5692; Thought Tracing: 0.3478), although closed-source models still lead on this benchmark (e.g., GPT-5-mini: 0.9121).

On social and affective evaluations, S^3-Chat maintains strong performance without sacrificing interaction quality. It achieves 4.23 on SOTOPIA, comparable to strong baselines (Qwen2.5-7B: 4.25; Mem0: 4.24), and reaches 71.17

Table 1. **Main results across personalization and social reasoning benchmarks.** Performance comparison of S^3-Chat against closed-source references, open-source baselines, and specialized SOTA methods (Qwen2.5-7B-Instruct as base model). Evaluations span long-term memory (PersonaMemV2, LongMemEval), theory-of-mind inference (BigToM, TOMi), and goal-driven social dynamics (Sotopia, EQBench, PrefEval). The best and second-best results among non-closed-source models are highlighted. Note: "-" indicates that the baseline configuration is incompatible with the specific benchmark settings.

Models	PersonaMemV2	LongMemEval	BigToM	TOMi	SOTOPIA	EQBench	PrefEval
Closed-source Models							
GPT-4o-mini	0.4856	0.1260	0.7987	0.2875	4.10	77.90	0.9360
GPT-5-mini	0.6146	0.1767	0.9121	0.4695	4.04	82.07	0.9820
Open-source Models							
Qwen2.5-7B-Instruct	0.5263	0.1500	0.7500	0.6827	**4.25**	**73.21**	0.8220
Llama-3.1-8B-Instruct	0.4308	0.1160	0.6538	0.1537	4.17	61.48	0.8260
Existing SOTA Methods							
Mem0	0.5025	0.1620	0.7188	0.6733	4.24	71.73	0.8400
Sotopia-π	0.2490	-	0.5692	0.3798	4.19	50.20	0.7941
AutoToM	-	0.1500	0.2677	**0.7823**	-	73.17	0.4780
Thought Tracing	-	-	0.3478	0.6600	-	68.80	0.8260
Ours							
S^3-Chat (w/ Qwen2.5-7B)	**0.5467**	0.1660	0.8700	0.6828	4.24	71.17	**0.9240**
S^3-Chat (w/ Llama-3-8B)	0.4093	**0.2804**	**0.9563**	0.1023	4.15	62.07	**0.9240**

on EQBENCH, close to Mem0 (71.73). Finally, on PREFEVAL, S^3-Chat achieves the best reported score (0.9240), surpassing open-source baselines and prior methods, indicating improved preference consistency and personalization. Taken together, these results demonstrate that S^3-Chat provides robust gains in long-term memory and ToM reasoning while preserving social and emotional competence.

Figure 4 demonstrates the high sample efficiency of the S^3-Gen pipeline. The positive correlation between data volume and preference adherence score suggests that the model effectively internalizes complex user dynamics through our privileged distillation paradigm.

Limitations. Our method shows relatively smaller gains on EQBench compared to other benchmarks, suggesting that emotional intelligence may require different training signals than those captured by our user state framework. Additionally, performance on the longest-context scenarios in LongMemEval remains challenging, indicating room for improvement in very-long-horizon memory tasks.

5. Conclusion

This paper addresses a key bottleneck for personalized LLMs: the lack of scalable interaction data that preserves the causal linkage between a user's latent state, intentions, and utterances. To this end, it proposes S^3 **(Structured State Simulation)**, which explicitly maintains an evolving user state during data synthesis to improve both short-term coherence and long-term consistency.

S^3 consists of a white-box simulator (S^3**-Sim**) with turn-level and event-level state updates, and a privileged-information distillation pipeline (S^3**-Gen**) that transfers oracle supervision into a deployable model that only observes dialogue history. Experiments show that the resulting S^3**-Chat** improves sample efficiency and yields consistent gains on long-horizon personalization tasks, with particularly strong improvements on **Theory-of-Mind** evaluations, indicating better intent and belief inference.

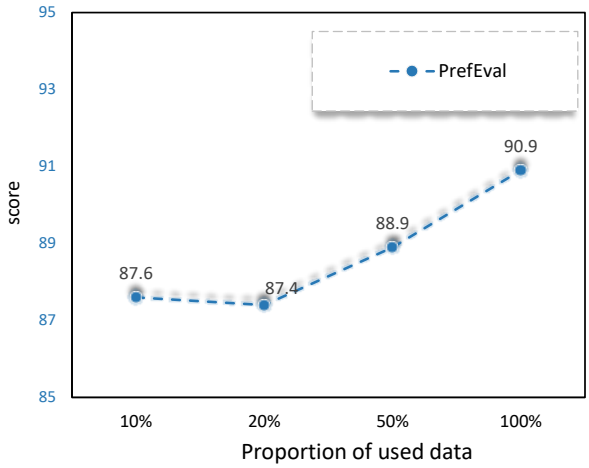


Figure 4. **Scaling analysis on PrefEval.** The performance of S^3-Chat improves consistently as the proportion of state-aware synthetic data increases.

To address remaining challenges in affective modeling and long-context personalization, future work should strengthen affect dynamics and state tracking while implementing safeguards against privacy risks and bias. Overall, S^3 provides a principled path toward controllable, scalable data generation for deeper personalization.

Impact Statement

The proposed framework for state-aware user simulation presents significant opportunities for advancing personalized artificial intelligence while introducing ethical challenges that require careful consideration. By synthesizing high-fidelity conversation trajectories grounded in latent user states, this research enables the development of agents that better understand human intent and theory-of-mind, which has direct beneficial applications in fields such as healthcare, education, and accessibility. In healthcare, it allows for more empathetic digital assistants that track a patient's emotional state over long-term interactions, while in education, it powers personalized tutors that adapt to specific learning styles through structured state tracking. However, the ability to simulate human behavior with high causal depth introduces risks related to privacy and surveillance, as techniques used to infer latent states could be repurposed to profile individuals more effectively. Furthermore, if the initial hierarchical JSON taxonomy contains historical social biases, the recursive update mechanism could inadvertently reinforce these stereotypes. Highly realistic simulators could also be misused by malicious actors to generate personalized misinformation campaigns. To mitigate these risks, we have integrated discriminative filtering and logical conflict detectors to reject biased scenario instantiations. We advocate for the establishment of community standards for watermarking synthetic data and encourage interdisciplinary collaborations with social scientists to ensure that simulated persona evolutions remain ethically aligned and psychologically realistic.

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A. Detailed Algorithmic Implementation

A.1. Formalization of the Behavioral Mode Policy

The determination of the transient behavioral mode $\mathcal{B}_t$ is governed by a latent decision function f_ϕ, implemented via a constrained prompting strategy that emulates a variance-maximizing prior. Formally, given the current latent user state s_t and dialogue history $H_{<t}$, the simulator selects a mode $b \in \mathcal{V}_b$ to maximize interaction diversity. This selection process is modeled as a categorical distribution conditioned on the hierarchical persona constraints:

$$P(\mathcal{B}t = b \mid s_t, H{<}\,t) = \text{LLM}(b \mid \text{Instr}(\mathcal{V}b), \mathcal{U}, \mathcal{C}, H{<}\,t) \tag{10}$$

To ensure the synthesized trajectories remain non-monotonic and avoid the passive compliance trap, the prompt configuration forces the underlying model to prioritize strategies that introduce epistemic challenges or social disclosures when the context permits. By delegating this choice to a high-capacity model acting as the policy engine, the framework captures the stochastic yet goal-oriented nature of human intent.

A.2. Conflict Detection and Backtracking Mechanism

The integrity of the stateful simulation is maintained through a two-stage validation protocol that acts as a supervisory layer over the raw LLM outputs. The conflict detector evaluates each generated utterance m_t against the structured JSON profile $\mathcal{U}$ to identify logical or persona-based inconsistencies.The first stage, Logical Alignment Verification, utilizes the LLM to perform an NLI-style (Natural Language Inference) check between the candidate utterance and the immutable traits in $\mathcal{U}$. If an entailment contradiction is detected (e.g., a trait-defined technophobe providing complex coding advice), the second stage, Backtracking Correction, is initiated. During backtracking, the simulator either regenerates the utterance with stricter attribute injection or, if the drift is deemed significant, triggers a macro-event E_t to synchronize the profile with the evolving narrative.

A.3. Implementation of Recursive Hierarchical Updates

The recursive evolution of the persona tree $\mathcal{U}$ addresses the challenge of long-term concept drift without sacrificing structural coherence. When an external event E occurs, the update propagates from the root to the leaves of the JSON taxonomy using a cascade-conditioning logic:

$$n.attr' \leftarrow \text{LLM}(n.attr \mid E, \text{Path}(root \rightarrow n)) \tag{11}$$

This ensures that any modification to a high-level attribute (e.g., "Career") automatically constrains the valid search space for subordinate attributes (e.g., "Daily Routine" or "Interests"). By utilizing the LLM as a recursive transformation engine, we maintain a causally consistent persona that evolves organically in response to external shocks, preventing the disjointed identity shifts common in flat-buffer memory architectures.

A.4. Note on Privileged Supervised Fine-tuning

In response to technical constraints regarding logit access from closed-source Oracle models, the distillation process is implemented as a hard-target supervised fine-tuning task under the Learning Using Privileged Information (LUPI) framework. The student model S^3-Chat is trained to approximate the Oracle's optimal decision boundary by minimizing the negative log-likelihood of the expert responses a_t. This objective effectively forces the student to learn an internal feature mapping that bridges the observable history $H_{<t}$ to the privileged user state s_t, achieving state-aware performance through implicit inference at runtime.